

Explainability Design

Explainability Overview =>

Explainability refers to the ability of the system to clearly communicate why a reputation value changed. The explanation must→

- Identify the rule that triggered the update.
- Describe the reason for the reputation change.
- Be understandable to a human user.

This model does not rely on artificial intelligence. All explanations are generated using deterministic rule-based logic.

Why Explainability Is Necessary =>

Blockchain systems provide transparency in terms of recorded data, but the reasoning behind state changes is not always immediately clear to users. Explainability ensures that reputation updates are not only visible, but also understandable.

An user can observe that a reputation value has changed, but may never understand→

- Why the reputation increased.
- Why the reputation decreased.
- How inactivity affected their reputation score.

Providing structured explanations improves→

- User trust in the system.
- Transparency of system behavior.
- Auditability of reputation updates.
- Clarity for users interacting with the reputation model.

Explanation Categories =>

The system defines a **limited** set of explanation categories. The current explanation types include:

- Initial Reputation Assigned
- Reputation Increased/Decreased Due to Interaction
- Reputation Decreased Due to Inactivity
- Reputation Reduced to Minimum Limit
- No Change (if applicable)

Explanation Generation Mechanism =>

Explanations are generated directly inside the smart contract when a reputation change happens. The process is these steps:

- The system evaluates which rule triggered the reputation update.
- The corresponding explanation type is selected.
- The user gets notified by the smart contract.

The notification includes →

- User address (id)
- Reputation change value (delta)
- Explanation message
- Timestamp of the update

For example →

Address: user id

Delta: -10

Reason: Reputation decreased due to inactivity

Timestamp: block time

These events become part of the immutable blockchain transaction log..

Event-Based Explanation Storage =>

Explanations are emitted as blockchain events rather than being permanently stored in contract state. This choice is made because:

- Events require lower gas cost compared to state storage.
- Events can be searched through blockchain logs.
- To preserve auditability without increasing on-chain storage.

***Deterministic* Explanation Mapping =>**

Each reputation rule in the system is associated with a predefined explanation. There is:

- No randomness in explanation generation.
- No hidden logic.
- No dependency on external data sources.

If the same conditions occur, the same explanation will always be emitted. This guarantees:

- Predictability
- Consistency
- Verifiability

Design Limitations =>

The explanation system is intentionally simple and rule-based. It does not:

- Generate complex natural language explanations.
- Interpret behavioral patterns beyond defined rules.